Rohan Batra

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Education

University of Virginia Charlottesville, VA

B.S. in Computer Science & B.A. in Mathematics, Major GPA: 3.9/4.0 *May 2028*

- Relevant Coursework: Data Structures and Algorithms, Computer Systems & Architecture, Software Development, Linear Algebra, Probability, Discrete Math, Differential Equations, & Calculus I-III
- Activities: Sports Analytics @ UVA, College Mentors for Kids, Wahoops, ML @ UVA, HooHacks, Google Developer Club, Claude Developer Club

Experience

Arcturus Technologies Jun. 2026 – Aug. 2026

Software Engineering Intern *Leesburg, VA*

- Developed a **binary classification prediction** model by leveraging the **Yahoo Finance API** & **SciKit-Learn**. Trained on **OHLCV** features using **logistic regression**, improving stock price prediction accuracy by **15%**.
- Built a **backtesting engine** using **Python, NumPy, Pandas, & Matplotlib**; Integrated **Monte Carlo simulations** to evaluate the performance of multiple trading strategy theses.
- Implemented comprehensive risk and statistical metrics, generating **Sharpe Ratios**, **drawdown profiles**, and **confidence intervals**, reducing strategy performance analysis time **to mere minutes**.

Mathnasium Aug. 2022 – Jun. 2025

Tutor *Leesburg, VA*

- Elevated academic performance for **200+** students in subjects up to **Calculus II** by identifying core knowledge gaps and breaking down complex problems into something intuitive and understandable for them.
- Maximized student success rates and retention by continuously assessing academic progress. Increased over 100 students' exam grades by **20+%** by using individualized teaching methods.

Projects

Course Spot Bot | GitHub | *Python, Google Cloud Platform*

- Eliminated missed course registrations by creating a serverless Python microservice that tracks specific classes/sections via the UVA **SIS REST API**.
- Delivered instant class availability alerts with **under 5-second** latency by parsing JSON payloads and a notification system.
- Achieved **24/7** autonomous system uptime and eliminated manual page refreshes by deploying the application via Google Cloud Run and Cloud Scheduler on a cron interval.

JSON Validator | GitHub | *C++23*

- Lightweight, robust **JSON validator** in **C++23** capable of parsing and verifying the structural integrity of JSON data strings against standard formatting rules.
- Implemented a **lexer and recursive parser** to accurately tokenize and validate JSON objects, arrays, and primitive data types, ensuring optimal execution performance.

Terminal Socket Chat Server | GitHub | *C, POSIX Sockets, TCP/IP*

- Built a terminal-based, full-duplex chat application in **C** that functions natively as either a client or host server.
- Handled standard input and network data by writing an event loop around the `poll()` system call, allowing for concurrent reading and writing without thread blocking.
- Implemented unbuffered I/O and strict connection state handling to ensure messages display instantly and connections terminate cleanly without memory leaks.

Music Agenda | GitHub | *Swift, SwiftUI, Apple Music API*

- Developed a native macOS application to catalog album progression & completion, successfully handling concurrent artist and album queries achieving minimal search latency via the **Apple Music API**.

Competitions & Programs

Estimathon – Participant Nov. 2026

- Placed **1st out of 25+** teams in **Jane Street**–sponsored Estimathon by collaborating solving **Fermi estimation** problems.

Akuna Capital Options 201 – Participant Oct. 2026

- Accepted into a selective virtual options course (**cohort of 70**) covering options pricing, volatility, and trading concepts; completed all quizzes with **strong scores**.

Jane Street Mystery Planet – Participant Jul. 2026

- Selected for an invite-only puzzle competition and market simulation at Jane Street's NYC office; **placed 4th**.
- Collaborated in a team to solve logic and pattern-recognition challenges; Developed trading strategies by reasoning about price discovery, market dynamics, and participant incentives.

Technical Skills

Languages: Python, Java, C/C++, HTML/CSS, SQL, MATLAB, LaTeX
Frameworks & Libraries: React, Node.js/Express, Pandas, NumPy, Matplotlib
Developer Tools: Git/GitHub, Google Cloud/Apps Scripts, Firebase, JUnit Testing, REST/WebSocket, CI/CD, Agentic AI/LLMs